

1. WAFT Menu of Options and Examples

Wave Agent Framework & Tools

A Python framework for directed evolution of self-modifying AI agents

“Don’t just build agents. Breed them.”

1.1. Core Commands

1.1.1. Project Management

waft new <name>

Creates a new evolutionary laboratory with uv project structure, `_pyrite` memory, and Empirica tracking.

waft verify

Verifies project structure and configuration.

waft evolve --agent <name>

Run evolutionary cycle: Spawn variants → Evaluate in Gym → Select fittest.

1.1.2. Documentation Generation

waft-docs field-guide

Generate field guide PDFs (layman, professional, scientist levels).

waft-docs booklet

Assemble complete booklet with all field guides.

waft-docs session-summary

Generate comprehensive session summary PDF.

waft-docs all

Generate everything: field guides, booklets, printer-friendly versions.

1.1.3. Evolution & Gym

waft spawn --agent <name> --mutation <file>

Create agent variant with genetic mutations.

waft eval --agent <name>

Evaluate agent fitness in Scint Gym (Reality Fracture Detection).

waft evolve --agent <name> --generations <n>

Run full evolutionary cycle for multiple generations.

1.1.4. Pantheon (Higher Beings)

Magistrate

God of Precedent and Body of Proof. Organizes proof cases into precedents.

Judge

God of Judgment and Evaluation. Evaluates claims against Body of Proof.

The Reasoner

God of Reasoning Traces. Maintains traceable reasoning chains.

GitHub God

God of Repository Management. Generates rollups and tracks operations.

Paperwork God

God of Paperwork and Documentation. Manages forms and bureaucratic processes.

Typist God

God of Typst Document Generation. Tracks compilations, templates, and usage statistics.

1.2. Key Features

1.2.1. The Three Pillars

1. The Substrate (Code as DNA)

Agents write their own Python source code. Every agent has a unique genome ID (SHA-256 hash). Mutations are code changes. Evolution is hot-swapping better genomes.

2. The Physics (Scint System)

Reality Fracture Detection acts as natural selection. Agents face quests testing ability to handle: SYNTAX_TEAR, LOGIC_FRACTURE, SAFETY_VOID, HALLUCINATION. Fitness measured by stability, efficiency, safety.

3. The Flight Recorder

Rigorous telemetry for phylogenetic trees. Every evolutionary action recorded with complete context: Genome ID, Parent ID, Generation, Event Type, Payload, Fitness Metrics.

1.3. Template System

1.3.1. Typst Templates

Available Templates:

- Appreciated Letter - Business/personal letters
- Brilliant CV - Professional CV/resume
- Worldbuild ISO - ISO 7010 safety symbols
- Worldbuild Quill - Quantum circuit diagrams
- Worldbuild Yagenda - Event schedules and agendas
- Flow Way - Academic papers
- Arkheion - Research papers
- Wonderous Book - Book formatting
- And many more...

Usage:

All templates auto-discovered via `TypstTemplateRegistry`. Use wrapper functions or direct `TypstCompiler`.

1.3.2. LaTeX Templates

Comprehensive LaTeX template library including academic papers, CVs, letters, and more. Integrated with WAFT's document generation system.

1.4. Examples & Showcases

1.4.1. Interactive Demos

`examples/interactive_demo.py`

Experience WAFT documenting itself in real-time with ASCII art and animations.

`examples/advanced_demo/advanced_demo.py`

Advanced demo showcasing self-documentation, PDF organization, and meta-cognitive integration.

examples/generate_waft_intro_one_pager.py

Generate one-pager introduction to WAFT using evolution tools.

1.4.2. Document Generation

scripts/generate_waft_docs.py

Unified CLI for generating field guides, booklets, and session summaries.

examples/generate_session_recap_pdf_waft.py

Generate session recap PDFs using ChatDistiller, StylingGenome, and TwoPageGenerator.

examples/generate_feature_showcase.py

Comprehensive feature showcase demonstrating all WAFT capabilities.

1.5. Integration Points

1.5.1. Empirica (Epistemic Tracking)

WAFT integrates with Empirica for epistemic self-assessment and learning tracking.

1.5.2. Karma System

Complete karma economy with KarmaMarket, KarmaCollector, and Afterlife Market (Treasure Tavern).

1.5.3. Being System

Timeful agents that learn skills, evolve, and pass memories upward through ancestral chains.

1.5.4. Reality System

Spin up simulation environments where beings can learn and evolve.

1.6. Quick Reference

Installation

`uv tool install waft`

New Project

`waft new my_lab`

Verify Setup

`waft verify`

Help

`waft help`

1.7. Additional Resources

1.7.1. Documentation

• README.md - Main project documentation • AGENTS.md - AI agent instructions • WAFT_CONTEXT_DUMP.md - Comprehensive system overview • docs/ - Detailed documentation directory

1.7.2. Work Efforts

• _work_efforts/ - Johnny Decimal organized work tracking • Uses work-efforts MCP server for management

1.7.3. Pantheon

• _pantheon/ - Higher Beings (Gods) system • Timeless Forces that Bind Reality Together

1.7.4. Realms

• _realms/ - Simulation environments • Bureaucracy Realm, PDFme Realm, and more

Generated: 2026-01-19

WAFT - Wave Agent Framework & Tools

“Don’t just build agents. Breed them.”